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Flood protection system

Abstract

A flood protection system (10) on a watercourse (16) with a flood-prone area (18) includes a main pipe (12) along the watercourse (16) in this flood-prone area (18). At least parts of the watercourse (16) are guided through the main pipe (12) in the direction of flow (28). A drive turbine (62) in the main pipe (12) increases the flow velocity of the flowing water (16) in the main pipe (12) in the event of flooding, the main pipe (12) flowing back into the flowing water (16).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of German Patent Application DE 10 2023 136 702.1, filed on Dec. 27, 2023, the content of which is incorporated by reference in its entirety.

BACKGROUND

(2) In many river regions, flooding occurs when the river floods. This causes considerable economic damage to national economies. Flood waters often reach cities and destroy large parts of residential and industrial facilities. Flooding can also cause enormous damage to agriculture if the agricultural land is not specifically designed for this purpose. For this reason, dikes or dams are built along the course of rivers to prevent flooding or, if necessary, retention areas are designated against the increased volumes of water. In the case of rivers that flow through cities, such flood protection is often difficult due to the large amount of land required. For this reason, special flood walls are erected at the edges of rivers during floods. Due to their construction, dykes and dams are at risk of softening and breaking. They often consist of piled up sand and/or gravel, which is held together by the roots of plants. Flood walls are built every time there is a flood to provide flood protection. This incurs costs for the state and local authorities.

(3) DE 102006001715 A1 describes a device for preventing water from overflowing from a river

during floods, which provides a rigid hollow construction that does not allow water to pass through and is placed along the bank of the river. The construction is made of pipe halves, with the lower pipe halves permanently placed on the bank of the river. The upper pipe halves have an outer diameter that corresponds to the inner diameter of the lower pipe halves. These constructions serve on the one hand as a wall and on the other hand flowing water can be guided through them.

(4) DE 10 2011 012 777 A1 relates to a device and a method for preventing flooding. Using a simple technical design, water is pumped from a flood-carrying river into a more distant flood basin with the aid of supply pipes and main pipes.

(5) DE 10 2013 109 763 A1 relates to flood protection on a watercourse with an area at risk of flooding, containing a pipe along the watercourse in this area at risk of flooding, whereby parts of the watercourse are guided through the pipe in the direction of flow. A drive turbine is described in the pipe, which increases the flow velocity of the watercourse in the pipe. Floods occur because more water volume arrives in a river than can flow out of the river bed. The flow velocity is too low for the cross-section. The water overflows the riverbed and then seeks its destructive path into the surrounding area. The water can also flow through a pipe parallel to the river bed. This pipe contains a drive turbine that increases the flow speed. This allows more water volume to be transported in a controlled manner with the same river cross-section, e.g. past a city.

(6) The known flood protection systems have the disadvantage that a flood basin must be provided to transport the water away from the flood-prone area. Furthermore, in normal conditions, the pipes carry the water unintentionally from the watercourse past the flood area. In warm periods with low water levels, the riverbed of the watercourse in the area of the flood zone can dry out quickly. This is particularly detrimental to the flora and fauna there.

SUMMARY

(7) The disclosure relates to a flood protection system on a watercourse with a flood-prone area, comprising a main pipe along the watercourse in this flood-prone area, at least parts of the watercourse being directed through the main pipe in the direction of flow, a drive turbine being provided in the pipe to increase the flow velocity of the watercourse in the event of flooding in the pipe, the main pipe opening back into the watercourse.

(8) The disclosed flood protection system avoids the disadvantages of the prior art and can also be used at normal water levels.

(9) This is achieved by providing at least one branch element with at least two pipe branches in a flood protection system on a watercourse with an area at risk of flooding of the type mentioned at the beginning, the first pipe branch leading into the main pipe and the second pipe branch leading into a secondary pipe, the branch element being designed with an adjustable shut-off device which shuts off or releases a pipe branch. In this way, it is possible to decide whether water is led past the area at risk of flooding or directly back into the watercourse. This can prevent a river bed from drying out at relatively low water levels because the water is routed through the pipes.

(10) An advantageous embodiment of the flood protection system on a watercourse with a flood-prone area is that the shut-off device has a gate valve that blocks the flow of the first pipe branch and/or the second pipe branch. There are several conceivable applications for the slide valve. On the one hand, the gate valve can block a branch. This means that water is no longer fed through the branch element. On the other hand, the gate valve can also be designed in such a way that it closes either one pipe or the other. This allows the water flow to be influenced. Either the water is diverted around the danger zone during high water or it flows directly back when the water level of the watercourse is normal.

(11) A preferred embodiment of the flood protection system is achieved by the shut-off device having a mechanical, electrical, hydraulic or pneumatic drive for actuating the shut-off or release process. The shut-off device can thus be automated. It is now not necessary to actuate the shut-off device manually. The shut-off device is preferably operated by a drive.

(12) In a particular embodiment of the flood protection system, a processor-controlled control unit

is provided for the shut-off device. The control unit is provided for the automation of the shut-off device. The control system works with a microprocessor that can be easily programmed for the main control tasks. If required, the control unit can also be operated remotely via radio/mobile radio or cable in remote mode. The control unit adjusts the shut-off device in a suitable manner, for example by controlling the drive for the gate valve.

(13) A further advantageous embodiment of the flood protection system is achieved by providing at least one water level sensor, which detects the water level of the flowing water and transmits it to the processor-controlled control system of the shut-off device for evaluation. To further increase the degree of automation, the control system can use the water level sensor to determine normal water levels or a flood. The shut-off device of the branch element can be automatically adjusted accordingly.

(14) Preferably, the control unit then activates the shut-off device at a threshold value for the water level so that it releases or blocks either the first pipe branch or the second pipe branch.

(15) In a particular embodiment of the flood protection system, artificial intelligence supports the control process. Here, the artificial intelligence can decide whether or not an emergency can actually arise due to flooding. The artificial intelligence can be trained with various scenarios. For example, artificial intelligence can be trained so that the bow wave of a ship does not trigger a flood emergency.

(16) In a further preferred embodiment of the flood protection system, at least one pipe of the second pipe branch is led back into the flowing water, whereby a generator is provided in the second pipe branch to generate electricity. If there is no flood water, the generator in the second pipe branch can be used to generate electricity. The electricity generated in this way can be fed into the public grid for sustainable power generation.

(17) Furthermore, a special design of the flood protection system on a watercourse can be achieved by providing a gradient for the second pipe branch. At a normal water level, the water is guided through the pipe until there is a sufficient gradient. The secondary pipe can then be returned to the watercourse. The kinetic energy of the returning water can be used to operate a generator to produce electricity.

(18) One aspect of the invention is that in the flood protection system, at least one of the pipes is now arranged on an outer meander edge of the watercourse. This is where the flow velocity of the water is greatest, so that the greatest effect can be achieved here, for example with regard to a generator. Due to the increased flow velocity, there are also no or only very few fish in this area that could be unintentionally sucked into the pipe.

(19) It has also proved to be an advantageous embodiment of the invention if an animal protection device is provided on the flood protection system, which prevents animals from entering the pipes. This prevents living creatures from being unintentionally guided through the pipes and coming to harm. For example, this can be achieved by a sieve-like or net-like construction.

(20) Examples of embodiments are explained in more detail below with reference to the accompanying drawings. The invention is not intended to be limited solely to these listed embodiments. They merely serve to explain the invention in more detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a first example of a flowing body of water on which a flood protection system is arranged with a branching element with two pipe branches.

(2) FIG. 2 shows an enlarged section of the branching element as shown in FIG. 1, with the shut-off device blocking the first pipe branch and releasing the second pipe branch.

(3) FIG. 3 shows an enlarged section of the branching element as shown in FIG. 1, with the shut-off

device blocking the second pipe branch and releasing the first pipe branch.

(4) FIG. 4 shows a section of the first pipe with a drive turbine as shown in FIG. 1.

(5) FIG. 5 shows a cross-section through the flood protection system according to FIG. 1 in the area of the second pipe junction.

(6) FIG. 6 shows a cross-section through the flood protection system FIG. 1 in the area of the drive turbine.

(7) FIG. 7 shows a second embodiment example for a flowing body of water on which a flood protection system is arranged with a branching element with two pipe branches.

DETAILED DESCRIPTION

(8) In FIG. 1, the reference sign **10** designates a flood protection system. The flood protection system **10** has a first pipe, the main pipe **12**, which is guided along the shore **14** of a watercourse **16**, such as a river, a stream or a canal. Buildings **20** of a village or town are preferably located in an area **18** at risk of flooding. When the watercourse floods, these buildings **20** are severely affected and can be considerably damaged by the overflowing water.

(9) The main pipe **12** therefore receives water **22** in the higher upper area **24** of the watercourse **16** with an inlet pipe end **26**. The direction of flow of the water **22** is indicated by the arrows **28** in the watercourse and by the arrows **30** in the main pipe **12**. The main pipe **12** is led past the flood-prone area **18** and opens with a diverting pipe end **32** in a lower area **33** of the watercourse **16**. The main pipe **12** forms a barricade against flooding and thus serves as flood protection itself. The main pipe **12** thus forms a bypass to the watercourse **16** in order to be able to move the water volumes more quickly around the flood-prone area **18**.

(10) In the upper area **24** of the flowing water **16**, a branch element **34** is inserted in the main pipe **12**. The branch element **34** diverts the flow direction **30** of the main pipe **12** into a second pipe, the secondary pipe **36**. In this way, the water **22** entering the main pipe can be discharged directly back into the watercourse **16** upstream of the flood-prone area **18**.

(11) The branch element **34** has an adjustable shut-off device **38**. The adjustable shut-off device **38** consists of a slide valve **40**, which optionally directs the flow through the main pipe **12** or the secondary pipe **36**. For this purpose, the shut-off device **38** blocks either a pipe branch **42** to the secondary pipe **36** or a pipe branch **44** to the main pipe **12**. In this embodiment example, the slide valve **40** is driven by an electric motor **46** and pivoted into the desired position.

(12) The electric motor **46** is actuated by a microprocessor-controlled control unit **48**. The control unit **48** receives a signal via a signal line **49** from a water level sensor **50**. The water level sensor **50** supplies the current water level of the flowing water **16** as permanently as possible. The control unit **48** continuously evaluates the water level. If a certain threshold value is exceeded, the control unit **48** is activated. The control unit **48** controls the shut-off device **38** via a control line **51** in such a way that the pipe branch **42** to the secondary pipe **36** is blocked by the slide valve **40**, as shown in FIG. 3. For this purpose, the electric motor **46** is activated, which closes the slide **40** in front of the pipe branch **42** to the secondary pipe **36**. The control unit **48** is equipped with artificial intelligence. The artificial intelligence is trained in such a way that it can decide whether a flood is actually present. For example, it can prevent the bow wave of a ship from being recognized as high water.

(13) If the water level sensor **50** signals that the water level is below the threshold value, the main pipe **12** is blocked accordingly with the slide valve **40**.

(14) At normal water levels, accelerated forwarding of the water around the flood-prone area **18** is not necessary. Therefore, in this case, the water is directed into the secondary pipe **36** with the branch element **34**, as shown in FIG. 2. The direction of flow of the secondary pipe **36** is indicated by arrows **52**. The secondary pipe **36** also discharges in the upper area **24** upstream of the flood-prone area **18**. A generator **54** is arranged in the secondary pipe **36**, which is driven by a turbine drive **56**. The secondary pipe **36** is therefore arranged in such a way that there is a sufficient gradient to generate sufficient kinetic energy for the generator **54**. The current from the generator **54** can, for example, supply the surrounding buildings **20** with electrical energy.

(15) An animal protection device **58** is located in front of or at the inlet pipe end **26**. This animal protection device **58** can, for example, be designed as a net or sieve **60**, which prevents animals or plants from unintentionally entering the main pipe **12**.

(16) In the event of high water, the branch element **34** is set so that only the main pipe **12** is passable. A drive turbine **62** draws the water **22** from the watercourse **16** into the main pipe **12**. The water **22** is accelerated by the drive turbine **62**, diverted around the flood-prone area **18** with increased flow velocity and reintroduced into the watercourse **16** in the lower area.

(17) FIG. 2 shows an enlarged section of the branching element **34** as shown in FIG. 1. Here, the shut-off device **38** blocks the pipe branch **44** to the main pipe **12**. The pipe branch **42** to the secondary pipe **36** is unblocked, as described above. The water **22** thus flows through the secondary pipe **36** in the direction of flow **52**, driving the generator **54** to generate electricity. This operating mode is used when the water level is normal. The main pipe **12** is not used in this position of the slide valve **40**. The slide valve **40** is adjusted via the electric motor **46**. The control unit **48** regulates the electric motor **46** and thus also the position of the slide valve **40**.

(18) FIG. 3 shows an enlarged section of the branching element **34** as shown in FIG. 1. Here, the shut-off device **38** blocks the pipe branch **42** to the secondary pipe **36**. The pipe branch **44** to the main pipe **12** is unblocked as described above. This operating mode is used during high water. The water **22** thus flows through the main pipe **12** in the direction of flow **30**. The secondary pipe **36** is not used in the present position of the slide valve **40**. The slide valve **40** is adjusted via the electric motor **46**. The control unit **48** regulates the electric motor **46** and thus also the position of the slide valve **40**.

(19) FIG. 4 shows a section of the main pipe **12** with the drive turbine **62** according to FIG. 1. The main pipe **12** of the flood protection system **10** is shown as a schematic sketch in a longitudinal section. The drive turbine **62** is provided inside the main pipe **12**. The drive turbine **62** contains turbine blades **64**, which are arranged rotationally symmetrically on a drive shaft **66**. The turbine blades **64** are designed in such a way that when the drive shaft **66** rotates, they accelerate the water flowing through in the direction of flow corresponding to the arrows **30**. Guide vanes **67** direct the water **22** onto the turbine blades **64** at an optimum angle, thus ensuring that the volume of water flowing through the main pipe **12** is considerably increased in a time interval due to the higher flow velocity. The flow velocity through the main pipe **12** can be adapted to the respective flood. The drive turbine **62** also operates in a known reverse function as an electrical generator **68**, which can feed its electrical energy into the power grid.

(20) FIG. 5 shows a cross-section through the flood protection system **10** according to FIG. 1 in the area of the pipe junction **42** to the secondary pipe **36**. The secondary pipe **36** is arranged in such a way that it has a gradient **70**. The water **22** flowing through the secondary pipe **36** can transfer its kinetic energy to the generator **54** via the turbine drive **56**. The water **22** then flows back into the watercourse **16**. This operating mode can be set primarily at a normal water level of the watercourse **16** with the branch element **34**.

(21) FIG. 6 shows a partial section of the watercourse **16** in cross-section. The watercourse **16** is located in its riverbed **71**. At high water, the water **22** flows over the shore **14** up to a dam **72**. The dashed line **74** indicates the water surface at high water. The solid line **76** in the riverbed **71** shows the water surface at normal water **76**. The water **22** only reaches the shore **14**. To prevent the dam **72** from also being flooded, the drive turbine **62** in the main pipe **12** can be switched on in the present embodiment example. As a result, in addition to the volume of water transported through the riverbed **71**, considerable quantities of water are also directed through the main pipe **12**. The amount of water regularly depends on the power of the drive turbine **62**. The higher the output, the more water **22** flows through the main pipe **12** of the flood protection system **10**. The main pipe **12** also serves as a stable wall for flood protection or to stabilize at least the dam **72**.

(22) The drive turbine **62** has the turbine blades **64**, which are arranged rotationally symmetrically around the drive shaft **66**. The turbine blades **64** force the water **22** through the main pipe **12** at

high pressure and thus at an increased speed.

(23) FIG. 7 shows the flood protection system **10**. The flood protection system **10** also has the main pipe **12**, which in this embodiment example is routed along the shortest possible route past the flood-prone area **18** with its buildings **20**. The buildings **20** are symbolic of a village, a small settlement or a town. When the watercourse floods, these buildings **20** are severely affected and can be significantly damaged by the overflowing water.

(24) In this embodiment example, the main pipe **12** therefore also receives the water **22** in the higher upper area **24** of the watercourse **16** with the inlet pipe end **26**. The inlet pipe end is located in the outer area **78** of a meander **80** of the watercourse **16**. Due to the high flow velocity of the watercourse **16**, practically no living creatures, in particular fish, are present in this outer area **78**. This can also make an animal protection device **58**, as shown in the embodiment example of FIG. 1, superfluous.

(25) The direction of flow of the water **22** is indicated by the arrows **28** in the watercourse **16** and by the arrows **30** in the main pipe **12**. The main pipe **12** is led past the flood-prone area **18** and flows into the lower area **33** of the watercourse **16** with the diverting pipe end **32**. The main pipe **12** thus forms a bypass to the watercourse **16** in order to be able to move the water faster around the flood-prone area **18**.

(26) In the upper section **24** of the watercourse **16**, the branch element **34** is inserted in the main pipe **12**. The branch element **34** diverts the flow direction **30** of the main pipe **12** into the secondary pipe **36**. In this way, the water **22** entering the main pipe **12** can be discharged directly back into the watercourse **16** upstream of the flood-prone area **18**.

(27) In accordance with the embodiment example of FIG. 1, the branching element **34** has the adjustable shut-off device with the slide valve **40**, which optionally directs the flow through the main pipe **12** or the secondary pipe **36**. For this purpose, the shut-off device **38** blocks either a pipe branch **42** to the secondary pipe **36** or a pipe branch **44** to the main pipe **12**. the mode of operation of the branching element **34** corresponds to the embodiment examples described above.

(28) At normal water levels, accelerated forwarding of the water around the flood-prone area **18** is not necessary. Therefore, in this case, the water is directed into the secondary pipe **36** with the branch element **34**, as shown in FIG. 2. The direction of flow of the secondary pipe **36** is indicated by arrows **52**. The secondary pipe **36** also opens in the upper area **24** upstream of the flood-prone area **18**. The generator **68** is arranged upstream of the branch element **34**. The generator **68** is driven by the turbine drive **62**. The secondary pipe **36** is designed in such a way that there is a sufficient gradient to generate sufficient kinetic energy for the generator **54**. The current from the generator **68** can, for example, supply the surrounding buildings **20** with electrical energy.

(29) In the event of high water, the branch element **34** is set so that only the main pipe **12** is passable. The generator **68** is then operated in reverse function as an electric motor, which drives the turbine drive **56**. The turbine drive **56** then draws the water **22** from the flowing water **16** into the main pipe **12**. The water **22** is accelerated by the drive turbine **62** and diverted around the flood-prone area **18** at an increased flow velocity and reintroduced into the watercourse **16** in the lower section.

LIST OF REFERENCE NUMBERS

(30) **10** Flood protection system **12** Main pipe **14** Shore **16** Watercourse **18** Flood-prone area **20** Buildings **22** Water **24** Upper area **26** Inlet pipe end **28** Flow direction in the watercourse **30** Flow direction in the main pipe **32** Diverting pipe end **33** lower area **34** Branch element **36** Secondary pipe **38** Shut-off device **40** Slide valve **42** Pipe branch to the secondary pipe **44** Pipe branch to the main pipe **46** Electric motor **48** Control unit **49** Signal line **50** Water level sensor **51** Control line **52** Flow direction of the secondary pipe **54** Generator **56** Turbine drive **58** Animal protection device **60** Sieve **62** Drive turbine **64** Turbine blades **66** Drive shaft **67** Guide vanes **68** Generator **70** Gradient **71** Riverbed **72** Dam **74** High water **76** Normal water **78** outer area **80** Meander

Claims

1. A flood protection system (10) on a watercourse (16) with a flood-prone area (18), comprising: a main pipe (12) arranged along the watercourse (16) in the flood-prone area (18), wherein at least a part of flowing water of the watercourse (16) is guided through the main pipe (12) in a direction of flow (28); a drive turbine (62) provided in the main pipe (12) for increasing a flow velocity of the part of flowing water in the main pipe (12) during flooding, wherein the main pipe (12) opens into the watercourse (16); at least one branching element (34) with at least two pipe branches (42, 44), wherein a first pipe branch (44) of the at least two pipe branches (42, 44) leads into the main pipe (12) and a second pipe branch (42) of the at least two pipe branches (42, 44) leads into a secondary pipe (36), wherein the branching element (34) is formed with an adjustable shut-off device (38) which shuts off or releases at least one of the at least two pipe branches (42, 44).
 2. The flood protection system (10) according to claim 1, wherein the adjustable shut-off device has a slide valve (40) which blocks the first pipe branch (42) and/or the second pipe branch (44).
 3. The flood protection system (10) according to claim 1, wherein the adjustable shut-off device (38) has a mechanical, electrical, hydraulic, or pneumatic drive for actuating a shut-off or release operation.
 4. The flood protection system (10) according to claim 1, wherein a processor-controlled control unit (48) is provided for the adjustable shut-off device (38).
 5. The flood protection system (10) according to claim 4, further comprising a water level sensor (50), wherein the water level sensor (50) detects a water level of the watercourse (16) and transmits the water level to the processor-controlled control unit (48) of the adjustable shut-off device (38) for evaluation.
 6. The flood protection system (10) according to claim 5, wherein the processor-controlled control unit (48) activates the adjustable shut-off device (38) at a threshold value for the water level and releases or blocks either the first pipe branch (42) or the second pipe branch (44).
 7. The flood protection system (10) according to claim 4, wherein an artificial intelligence supports a control process of the processor-controlled control unit (48).
 8. The flood protection system (10) according to claim 1, wherein the secondary pipe (36) of the second pipe branch (42) is led back into the watercourse (16), and wherein a generator (54) for generating electricity is provided in the second pipe branch (42).
 9. The flood protection system (10) according to claim 8, wherein a gradient is provided for the second pipe branch (42).
 10. The flood protection system (10) according to claim 1, wherein at least one of the main pipe (12) or the secondary pipe (36) is arranged on an outer area (78) of a meander (80) of the watercourse (16).
 11. The flood protection system (10) according to claim 1, further comprising an animal protection device (58) which prevents entry of animals into the main pipe (12).
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